MIRROR WORLD · TEAM PIGEONS

Locating a drone is hard.

A receiver hears a signal for a few seconds, a camera catches a few pixels, and the pieces don’t quite line up. Merging them takes a trained model.

And good, faithful training data barely exists.

What such a model learns from:

signal strength | signal type | timestamps | images | labelled boxes | **+ the truth behind every record**

OUR ANSWER

Simulate the world. Measure it. Export it.

Faithful by construction: every value comes from terrain and physics.

Simulated world

- Real terrain, 4 areas
- 4 drone types
- Real flight limits, wind

Measured world

- Radio, radar, cameras
- Microphones, Remote ID
- Fixes with their error

Export

- ZIP: sensors + truth
- TAK Cursor on Target
- Same seed, same bytes

NEXT

Advanced simulation
wind and weather by height, land cover, jamming

More detector types
satellite imagery, more radar and camera types

Action recommendations
where to place the next sensor, how to respond

LIVE DEMO · 60 SECONDS

Plan · Simulate · Export

1 **Plan: pick an area, place receivers, draw drone paths**

set the drone mix, wind and mast height, then run

`localhost:8765/planner/`

2 **Simulate: watch the sensors at work**

radio, radar, cameras and microphones, fixes with their error, the 3D heatmap

`localhost:8765/viewer/`

3 **Export: one click, the files behind it**

a ZIP with the truth, every sensor's records and the TAK events

`Export data (.zip) · out/<run>/`